



Project 3

GROUP C

Dynamic Finite Element Analysis of Structure ME-473

Authors:

FEDERICO CAZZULANI (386924),
KILLIAN CHARMILLOT (339817),
EMIL FOSSDAHL HECHT-NIELSEN (403992),
ROBERT KNOBLOCH (404031),
ADAM LECROART (345348)

Professors:

STEFANO BURZIO

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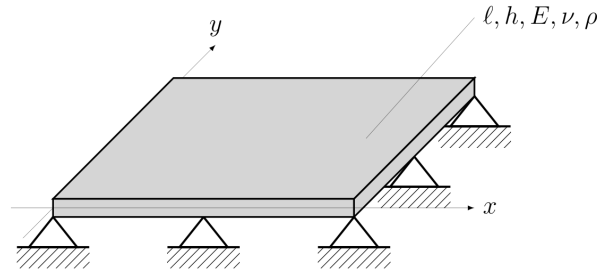


Figure 1: A square plate simply supported on all edges.

1 Introduction

For the third and final mini-project, a simply supported plate is given whose natural frequencies and mode shapes are to be analyzed.

Figure 1 shows the following setup: A square plate of side length l is simply supported along all its edges. It is made up of isotropic material.

The geometric and material parameters are given as follow:

- length of edge $l = 0.5\text{m}$
- thickness $h = 0.005\text{m}$
- elasticity $E = 210\text{GPa}$
- Poisson's ratio $\nu = 0.3$
- density $\rho = 7850\text{kg/m}^3$.

To solve the problem, this plate is modeled using both Allman-McLeod-Clough (AMC) and Crouzeix-Raviart (CR) plate bending elements. Their natural frequencies and corresponding mode shapes are calculated using the FEM with 2×2 , 4×4 , 8×8 and 16×16 elements and compared to the analytical solutions."

Additionally, the models are also applied to the same problem with different length to thickness ratios of the plate, comparing the performance of both models with the analytical thick plate solution, provided by Mindlin-Reissner plate theory, in these scenarios. It is investigated whether these thin plate models still hold under these conditions.

2 Methodology

Models have been developed specifically to calculate the mechanical behavior of plates. A plate is a planar structure that is only subject to transversal loads, and whose thickness is much smaller than the lengths of its sides.

2.1 Kirchhoff-Love Plate Bending Model

Two very important plate models are the Kirchhoff-Love and Mindlin-Reissner models. Compared to the latter, Kirchhoff-Love plates neglect shear forces, simplifying the calculations. This type of plate model is used in this FEM analysis.

2.1.1 Assumptions

The following assumptions are introduced in addition to the thin-plate assumption:

- The transverse fibers are inextensible, meaning the thickness is constant under every load ($\varepsilon_{33} = 0$),
- The material is homogeneous and linear-elastic,
- The displacement is small compared to the thickness,
- Load forces act only perpendicular to the plane and bending moment vectors lie in the plane,
- The shear strains ε_{13} and ε_{23} are neglected, meaning straight fibers normal to the plane are not bend/sheared.

2.1.2 Strong Form

Using the thin-plate and small-displacement assumptions, the general displacement vector can be broken down to only the vertical displacement u_3 :

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} \varphi_2 z \\ -\varphi_1 z \\ u_3 \end{bmatrix} = \begin{bmatrix} -z \frac{\partial u_3}{\partial x} \\ -z \frac{\partial u_3}{\partial y} \\ u_3 \end{bmatrix} \quad (1)$$

where φ_1 and φ_2 correspond to the rotation angles in the y-z and x-z plane respectively, and z corresponds to the distance along the thickness direction from a material point in the plate to the mid-surface in the undeformed configuration.

Since we get from the assumptions that, of the material strains, only ε_{11} , ε_{22} and $\varepsilon_{12} = \frac{1}{2}\gamma_{12}$ are nonzero, the strain-displacement relation simplifies to:

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix} = -z \underbrace{\begin{bmatrix} \partial_{xx}^2 \\ \partial_{yy}^2 \\ 2\partial_{xy}^2 \end{bmatrix}}_{\nabla_k} u_3 \quad (2)$$

Assuming isotropic material, the stiffness matrix reads:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \underbrace{\frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}}_{\mathbf{C}} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \gamma_{12} \end{bmatrix} \quad (3)$$

Since no external loads are applied in the problem, their treatment can be left out for now.

The moments acting on an infinitesimal element come down to

$$\mathbf{M} = \begin{bmatrix} M_{11} \\ M_{22} \\ M_{12} \end{bmatrix} = -\frac{h^3}{12} \mathbf{C} \nabla_k u_3 \quad (4)$$

Applying force and moment balance on that element, we get the strong form of the model and thus equations for the transverse displacement $u_3 \in C^4(\Omega \times [0, T])$:

$$\begin{aligned} \frac{h^3}{12} \nabla_k^T \mathbf{C} \nabla_k u_3 + \rho h \ddot{u}_3 &= 0 & \text{on } \Omega \times]0, T[\\ u_3 &= 0 & \text{on } \partial\Omega \times]0, T[\\ \mathbf{M}_n &= 0 & \text{on } \partial\Omega \times]0, T[\\ u_3(\cdot, 0) &= u_0 & \text{on } \Omega \\ \dot{u}_3(\cdot, 0) &= v_0 & \text{on } \Omega \end{aligned} \quad (5)$$

where ∇_k is the vector of partial derivatives given in (2) and $\mathbf{M}_n$ is the normal bending moment. In the case of a square plate: $\Omega = [-a, a] \times [-a, a]$. Equations 2 and 3 in (5) are the boundary conditions (simply supported plate), equations 4 and 5 the initial conditions of the problem.

2.1.3 Weak Form

To derive the weak form we introduce the virtual displacement δu and impose that

$$\begin{aligned} u_3 &\in U = \{u_3(\cdot, t) \in H^2(\Omega) | u_3 = 0 \text{ in } \partial\Omega \times]0, T[\} \\ \delta u_3 &\in V = \{\delta u_3 \in H^2(\Omega) | \delta u_3 = 0 \text{ in } \partial\Omega \} \end{aligned} \quad (6)$$

with the Sobolev space $H^2(\Omega)$. Multiplying equation 1 in (5) by δu and integrating over Ω one obtains:

$$\int_{\Omega} \delta u_3 \frac{h^3}{12} \nabla_k^T \mathbf{C} \nabla_k u_3 d\Omega + \int_{\Omega} \delta u_3 \rho h \ddot{u}_3 d\Omega = 0 \quad \forall \delta u_3 \quad (7)$$

The divergence theorem is applied to these integrals and the boundary conditions are imposed, leading to:

$$\frac{h^3}{12} \int_{\Omega} \nabla_k u_3 \mathbf{C} \nabla_k \delta u_3 d\Omega + \int_{\Omega} \rho h \ddot{u}_3 \delta u_3 d\Omega = 0 \quad (8)$$

Using the same approach on the initial conditions yields:

$$\begin{aligned} \int_{\Omega} \rho h \delta u_3 u_3|_{t=0} d\Omega &= \int_{\Omega} \rho h \delta u_3 u_0 d\Omega, \\ \int_{\Omega} \rho h \delta u_3 \dot{u}_3|_{t=0} d\Omega &= \int_{\Omega} \rho h \delta u_3 v_0 d\Omega, \end{aligned} \quad (9)$$

2.2 Galerkin Method

Using the Galerkin method, we approximate the (virtual) displacement using shape functions $\mathbf{H}(x)$ and a time dependent function $\vec{q}(t)$:

$$\begin{aligned} u_3^h(x, t) &= \mathbf{H}(x)\vec{q}(t) \\ \delta u_3^h(x) &= \mathbf{H}(x)\delta\vec{q} \end{aligned} \quad (10)$$

Inserting the approximated displacements into the weak form equation yields the semi-discrete form for a plate element:

$${}^e\delta\vec{q}^T \left(\int_{e\Omega} \frac{h^3}{12} (\nabla_k {}^e\mathbf{H})^T \nabla_k {}^e\mathbf{H} d\Omega \right) {}^e\vec{q} + {}^e\delta\vec{q}^T \left(\int_{e\Omega} \rho h {}^e\mathbf{H}^T {}^e\mathbf{H} d\Omega \right) {}^e\vec{q} = 0 \quad (11)$$

If we assume the mesh to be regular and all plate elements to be similar squares, the master element shape function can be used directly for all elements. Each plate element of size $a \times a$ is transformed to the master element, a square with coordinates ξ_1 and ξ_2 going from -1 to 1 . This simplifies the semi-discrete equation to

$${}^e\delta\vec{q}^T {}^a\mathbf{K} {}^e\vec{q} + {}^e\delta\vec{q}^T {}^a\mathbf{M} {}^e\vec{q} = 0 \quad (12)$$

with

$$\begin{aligned} {}^a\mathbf{B} &= \nabla_k {}^a\mathbf{H} \\ {}^e\mathbf{K} &= {}^a\mathbf{K} = \frac{h^3 a^2}{12} \int_{-1}^1 \int_{-1}^1 {}^a\mathbf{B}^T {}^a\mathbf{C} {}^a\mathbf{B} d\xi_1 d\xi_2 \\ {}^e\mathbf{M} &= {}^a\mathbf{M} = \rho h a^2 \int_{-1}^1 \int_{-1}^1 {}^a\mathbf{H}^T {}^a\mathbf{H} d\xi_1 d\xi_2 \end{aligned} \quad (13)$$

There are now two ways to proceed that are regarded in this project, using the plate element models Allman-McLeod-Clough (AMC) or Crouzeix-Raviart (CR). The difference between these two are the amount of degrees of freedom for each element and the shape function ${}^a\mathbf{H}$.

2.2.1 Allman-McLeod-Clough Element

For AMC elements, three degrees of freedom are assumed per node, the vertical displacement d and the two rotations θ_1 and θ_2 around the ξ_1 and ξ_2 axes:

$${}^e\mathbf{q}^i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\boldsymbol{\xi}^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \\ -\partial_{\xi_1} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \end{bmatrix} \quad (14)$$

The displacement u_3 is approximated using a 12-term polynomial, complete cubic polynomial, augmented with two (*geometrically invariant*) quartic terms, to comply with the 12 degrees of

freedoms per element:

$$\begin{aligned} {}^e u_3^h(x_1, x_2, t) = & a_1 + a_2 x_1 + a_3 x_2 + a_4 x_1^2 + a_5 x_1 x_2 + a_6 x_2^2 \\ & + a_7 x_1^3 + a_8 x_1^2 x_2 + a_9 x_1 x_2^2 + a_{10} x_2^3 + a_{11} x_1^3 x_2 + a_{12} x_1 x_2^3 \end{aligned} \quad (15)$$

With equation (14) one gains the following shape functions:

$$\begin{aligned} {}^a \mathbf{H} &= [{}^a \mathbf{h}_1, {}^a \mathbf{h}_2, {}^a \mathbf{h}_3, {}^a \mathbf{h}_4] \\ {}^a \mathbf{h}_i(\boldsymbol{\xi}) &= \begin{bmatrix} (1 + \xi_1^i \xi_1)(1 + \xi_2^i \xi_2)(2 + \xi_1^i \xi_1 + \xi_2^i \xi_2 - \xi_1^2 - \xi_2^2)/8 \\ a(1 + \xi_1^i \xi_1)(\xi_2^i + \xi_2)(\xi_2^2 - 1)/8 \\ -a(\xi_1^i + \xi_1)(\xi_1^2 - 1)(1 + \xi_2^i \xi_2)/8 \end{bmatrix}^T \end{aligned} \quad (16)$$

where $[\xi_1^i, \xi_2^i] \in \{[-1, -1], [1, -1], [1, 1], [-1, 1]\}$ are the local coordinates of node i .

The AMC element is not conforming, meaning it does not satisfy C^1 continuity.

2.2.2 Crouzeix-Raviart Element

CR elements have four degrees of freedom per node, adding the twist θ_{12} to the AMC element's DOF:

$${}^e \mathbf{q}^i(t) = \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \\ {}^e \theta_{12}^i(t) \end{bmatrix} = \begin{bmatrix} {}^e u_3^h(\boldsymbol{\xi}^i, t) \\ \partial_{\xi_2} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \\ -\partial_{\xi_1} {}^e u_3^h(\boldsymbol{\xi}^i, t)/a \\ \partial_{\xi_1 \xi_2}^2 {}^e u_3^h(\boldsymbol{\xi}^i, t)/a^2 \end{bmatrix} \quad (17)$$

In this case, the displacement is described by a 16-term polynomial of complete cubic order, augmented with three quartic, two quintic, and one sextic terms:

$$\begin{aligned} {}^e u_3^h = & a_1 + a_2 x_1 + a_3 x_2 + a_4 x_1^2 + a_5 x_1 x_2 + a_6 x_2^2 + \\ & + a_7 x_1^3 + a_8 x_1^2 x_2 + a_9 x_1 x_2^2 + a_{10} x_2^3 + \\ & + a_{11} x_1^3 x_2 + a_{12} x_1 x_2^3 + a_{13} x_1^2 x_2^2 + a_{14} x_1^3 x_2^2 + a_{15} x_1^2 x_2^3 + a_{16} x_1^3 x_2^3 \end{aligned} \quad (18)$$

The shape functions are more complex than for AMC elements, stemming from the additional DOF:

$$\begin{aligned} {}^a \mathbf{H} &= [{}^a \mathbf{h}_1, {}^a \mathbf{h}_2, {}^a \mathbf{h}_3, {}^a \mathbf{h}_4] \\ f_i(\xi) &= (-\xi^i \xi^3 + 3\xi^i \xi + 2)/4 \\ g_i(\xi) &= (\xi^3 + \xi^i \xi^2 - \xi - \xi^i)/4 \\ {}^a \mathbf{h}_i(\boldsymbol{\xi}) &= \begin{bmatrix} f_i(\xi_1) f_i(\xi_2) \\ a f_i(\xi_1) g_i(\xi_2) \\ -a g_i(\xi_1) f_i(\xi_2) \\ a^2 g_i(\xi_1) g_i(\xi_2) \end{bmatrix}^T \end{aligned} \quad (19)$$

In contrast to the AMC element, the CR element is conforming and thus satisfies C^1 continuity.

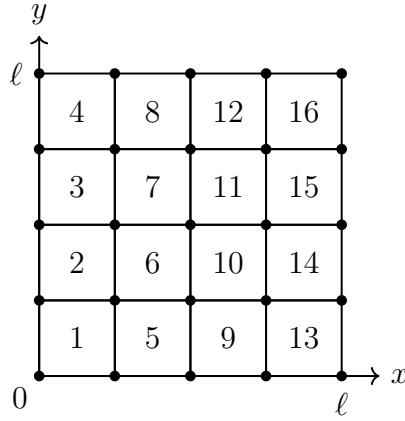


Figure 2: Meshing example on a 4x4 grid. The vertices are numbered in the same schema, starting from the bottom left.

2.3 Meshing

The plate is meshed regularly into $n \times n$ squares with side lengths ℓ/n . The numbering of the elements as well as the nodes start in the lower left, going up before continuing with the next column to the right. The numbering for a 4x4 grid is shown in figure 2.

The nodes of the elements are locally numbered starting from the bottom left going counter-clockwise.

From this the connectivity matrix is created.

2.4 Assembly

Next, the $3n \times 3n$ (AMC) or $4n \times 4n$ (CR) global stiffness and mass matrices are created respectively, where n denotes the number of nodes. The values of the element stiffness matrices ${}^e\mathbf{K} = {}^a\mathbf{K}$ and mass matrices ${}^e\mathbf{M} = {}^a\mathbf{M}$ are inserted into the global matrices at the right positions according to the connectivity table, yielding:

$$\begin{aligned}\mathbf{K} &= \mathbf{A}_e {}^e\mathbf{K} \\ \mathbf{M} &= \mathbf{A}_e {}^e\mathbf{M}\end{aligned}\tag{20}$$

The rows and columns relating to constrained nodes, meaning nodes whose displacement is imposed to be zero, are removed, leaving reduced matrices $\mathbf{K}_f, \mathbf{M}_f$ of free nodes. This yields the final equation for the displacement of the free nodes of the plate:

$$\mathbf{M}_f \ddot{\vec{q}}_f + \mathbf{K}_f \vec{q}_f = \vec{0}\tag{21}$$

2.5 Dynamic Analysis

To analyze the dynamic behavior of the system, we first solve the eigenvalue problem resulting from the ansatz $\vec{q}_f = \Phi e^{i\omega t}$.

$$(\mathbf{K}_f - \omega^2 \mathbf{M}_f) \boldsymbol{\Phi} = \mathbf{0} \quad (22)$$

Solving for the eigenvalues ω^2 and the resulting natural frequencies $f = \frac{\omega}{2\pi}$, we gain a first insight into the dynamic behavior of the system. The eigenvectors contained in $\boldsymbol{\Phi}$ are the corresponding mode shapes.

3 Results

Table 1 presents the four lowest natural frequencies for the AMC-element across different mesh sizes, while table 2 displays the same results for the CR-element. Figure 3 illustrates the corresponding mode shapes for 16x16 elements. The analytical frequencies listed in both tables are determined using equation (D.27) in [1]:

$$f_{i,j} = \frac{\pi}{2} \frac{i^2 + j^2}{\ell^2} \sqrt{\frac{D}{\rho h}}, \quad \text{with } D = \frac{Eh^3}{12(1 - \nu^2)} \quad (23)$$

While the analytical mode shapes for thin plate theory, are given by:

$$\phi_{ij}(x, y) = \sin\left(\frac{i\pi x}{\ell}\right) \sin\left(\frac{j\pi y}{\ell}\right), \quad \text{with } (x, y) \in [0, \ell] \times [0, \ell]. \quad (24)$$

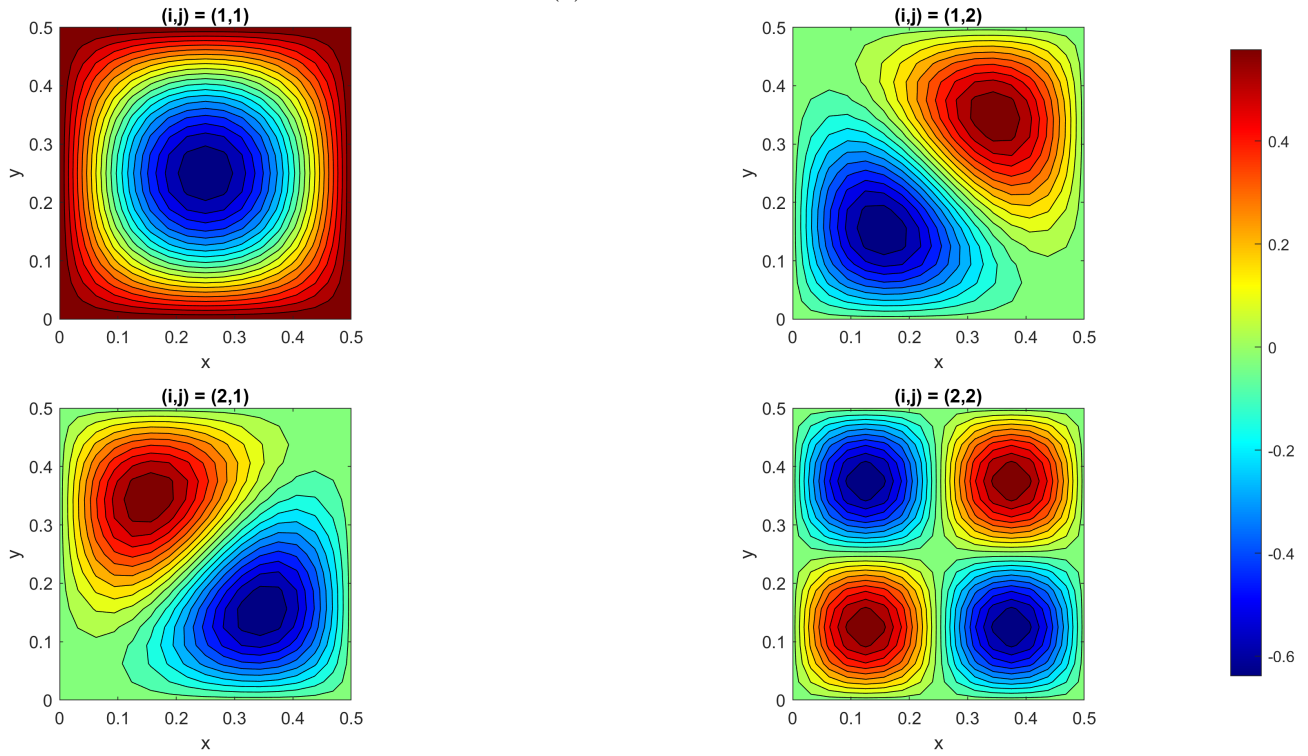
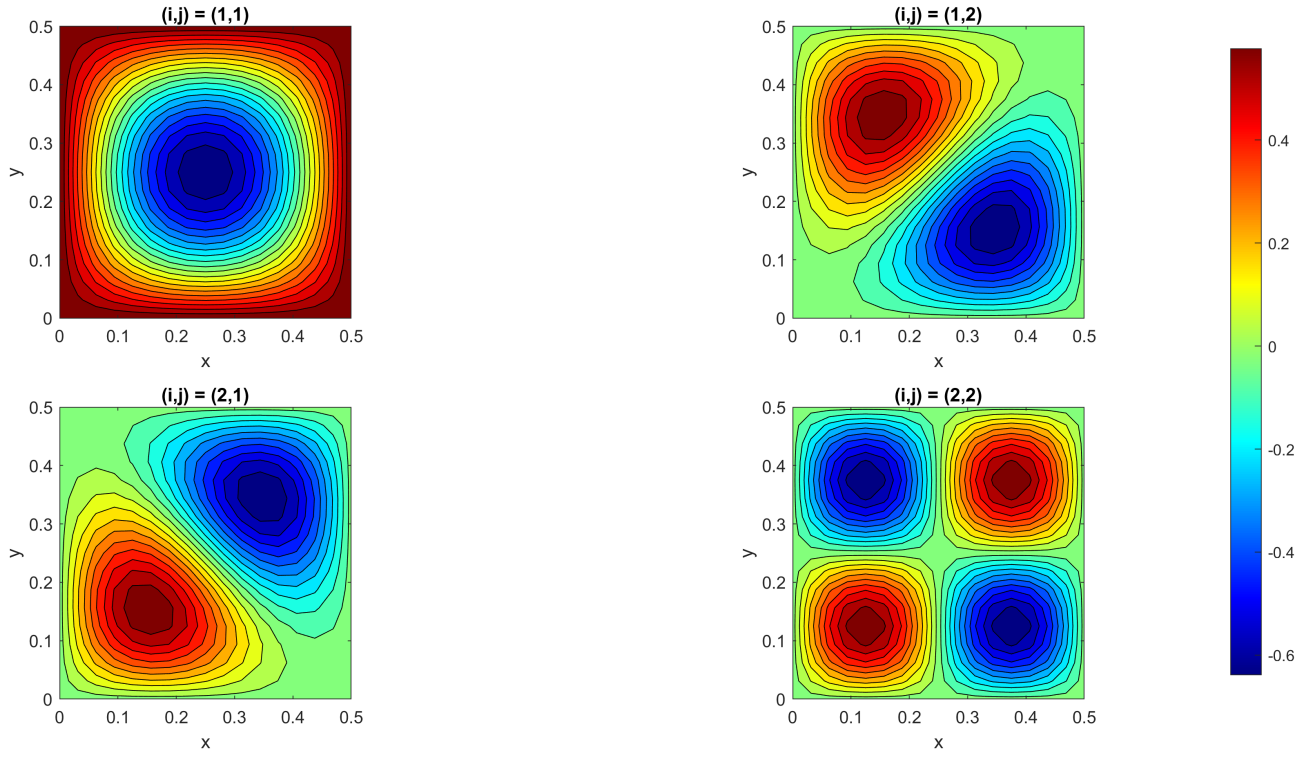
Table 1: Comparison of analytical frequencies and frequencies obtained with $n \times n$ AMC elements.

Index (i, j)	Analytical frequencies [Hz]	Frequencies for AMC element [Hz]			
		$n = 2$	$n = 4$	$n = 8$	$n = 16$
(1, 1)	98.343	89.7840	95.4066	97.5447	98.1391
(1, 2)	245.857	253.648	236.176	242.82	245.052
(2, 1)	245.857	253.648	236.176	242.82	245.052
(2, 2)	393.372	540.871	359.136	381.627	390.179

Table 2: Comparison of analytical frequencies and frequencies obtained with with $n \times n$ CR elements.

Index (i, j)	Analytical frequencies [Hz]	Frequencies for CR element [Hz]			
		$n = 2$	$n = 4$	$n = 8$	$n = 16$
(1, 1)	98.343	98.5550	98.3561	98.3438	98.3430
(1, 2)	245.857	264.006	246.496	245.899	245.860
(2, 1)	245.857	264.006	246.496	245.899	245.860
(2, 2)	393.372	418.023	394.220	393.424	393.375

Results obtained for error analysis and increasing thickness are presented in the discussion.

Figure 3: Mode shape contour plots relative to i, j for 16×16 elements

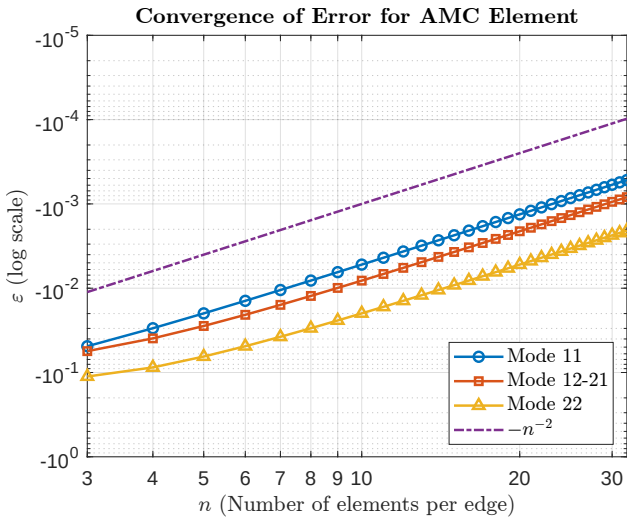
4 Discussion

For the simulation of simply supported thin plates, different finite element analyses were carried out by varying both the mesh density and the element type (AMC and CR). Focusing on the first fundamental mode shape and corresponding frequency, it is possible to observe the convergence behavior of the two formulations.

To quantify the deviation between the analytical modal frequencies f_{ij} and the estimated frequencies from the finite element analysis, $\tilde{f}_{ij}$, the relative error ε_{ij} is defined as:

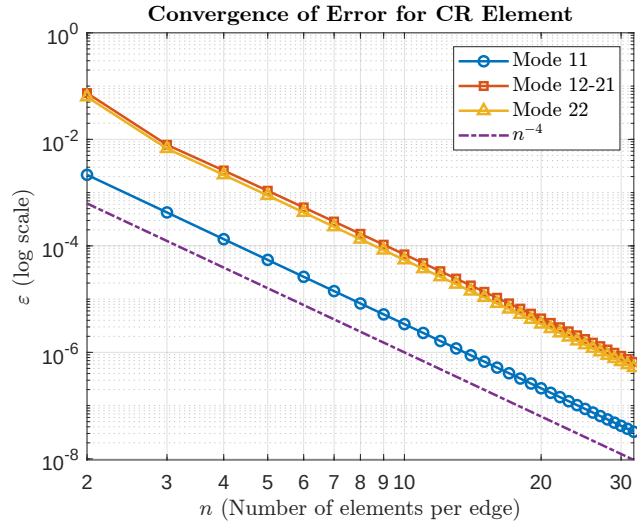
$$\varepsilon_{ij} = \left(\frac{\tilde{f}_{ij} - f_{ij}}{f_{ij}} \right) \quad (25)$$

It is important to note that the absolute value is intentionally not applied to the error expression, in order to preserve the sign and thereby reveal the convergence direction of each element type. The results are hereby reported:



(a) Convergence for **AMC elements type**

Note: No value is reported for $n = 2$, because of the different sign



(b) Convergence for **CR elements type**

Both element types achieve a good approximation of the first modal frequency (with relative error below 1%) using relatively coarse meshes: AMC elements converge at a mesh size of 8×8 , while CR elements already reach this accuracy at 2×2 . Nonetheless, the convergence behavior differs significantly between the two. AMC elements exhibit convergence from below, a behavior that is somewhat uncommon in finite element simulations. The error analysis suggests a convergence rate of approximately 2 for AMC elements.

In contrast, CR elements show a more typical convergence pattern from above and demonstrate a faster convergence rate of approximately 4. This allows CR elements to reach accurate results with fewer elements than AMC. This peculiar behavior highlights that the convergence rate of natural frequencies depends not only on the degree of the polynomial shape functions but also on their ability to accurately model the underlying physics of the problem. In fact, although

AMC elements employ a complete polynomial basis of degree 3, like the other element types, they exhibit a lower convergence rate because of the nonconformity of the shape function. Indeed, for this sort of element type, convergence rate is $\mathcal{O}(h^{p-1})$, where p represents the order of the higher completed polynomial base, and h the characteristic element size ($h \propto 1/n_{elements}$, with $n_{elements}$ being the total number of element in the mesh). On the contrary, despite having a complete polynomial space of degree 3, CR type includes additional higher-order terms, up to degree 6, which more effectively capture the curvature and energy characteristics of Kirchhoff-Love plates, leading to significantly better accuracy and higher convergence rates, which is effectively higher than the theoretical one from the degree of the shape function $\mathcal{O}(h^{p+1})$. In particular, for conformal elements, a priori estimations for eigenvalues and eigenvectors can be provided using Rayleigh and Courant-Fischer, asymptotic error estimation [2]:

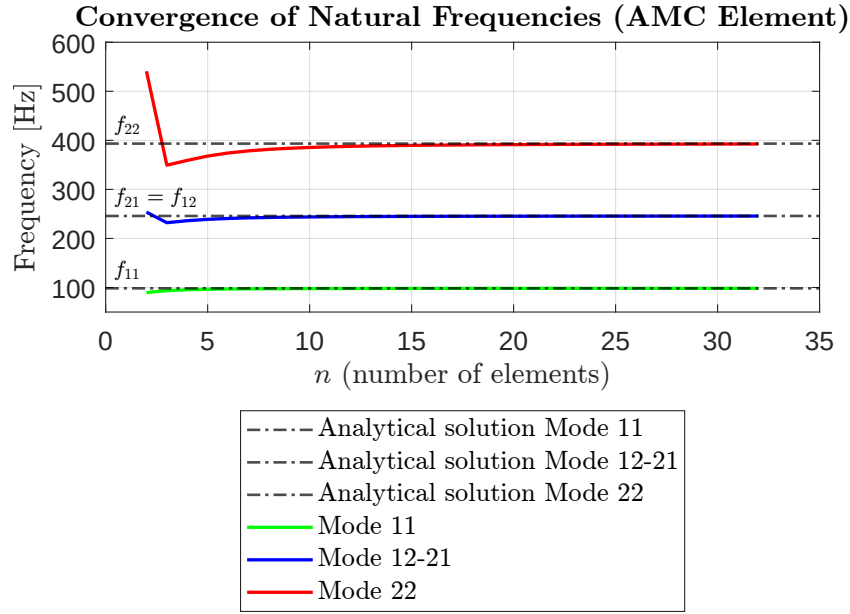
$$\begin{aligned}\lambda_i &\leq \lambda_i^h \leq ch^{2(m-k+1)}\lambda_i^{m+1} \\ \omega_i &\leq \omega_i^h \leq \bar{c}h^{2(m-k+1)}\omega_i^{2m+1}\end{aligned}\tag{26}$$

Where λ_i^h and ω_i^h are the approximated eigenvalues and frequencies, λ_i and ω_i are the exact ones. h represents the characteristic mesh size, c and $\bar{c}$ are constants independent to the mesh size. m is the degree of the highest complete polynomial used, being $m = 3$ for CR elements, and k denotes the highest derivative order appearing in the weak form, $k = 2$ for CR elements. Thus:

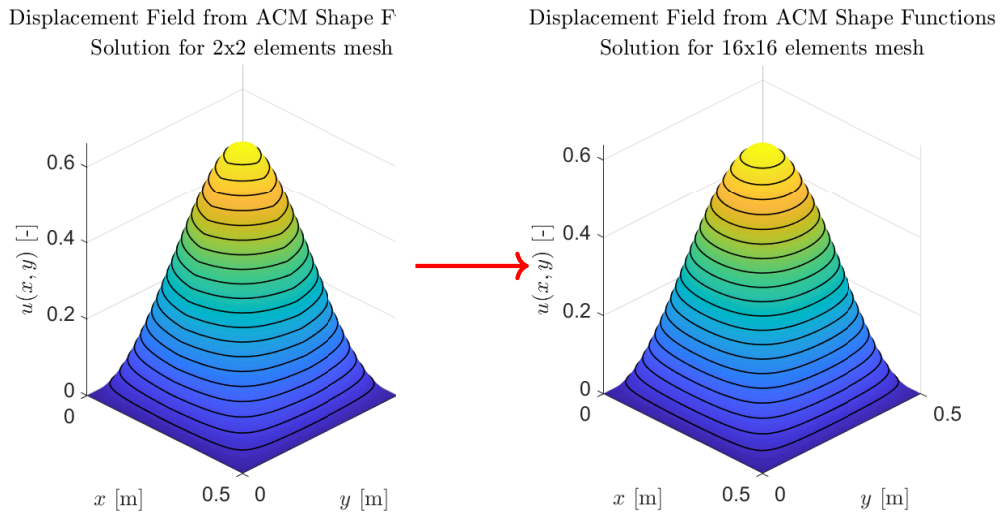
$$\omega_i \leq \omega_i^h \leq \bar{c}h^{2(3-2+1)}\omega_i^{2m+1} \propto h^4\tag{27}$$

Theoretically justifying the observed convergence from above, with rate h^4 for CR conforming elements.

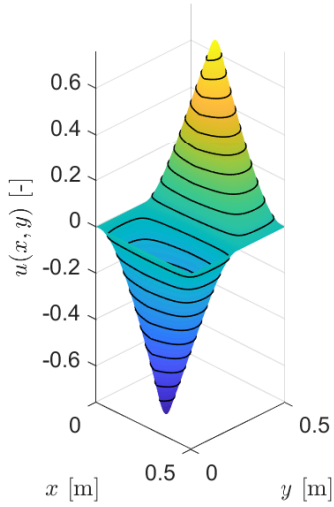
A particularly interesting behavior is observed in the AMC simulations for higher modal frequencies, namely f_{12} and its symmetric counterpart f_{21} (note that since they are equal, they are plotted only once), as well as f_{22} . For coarse meshes, especially with a 2×2 discretization, these frequencies are significantly overestimated compared to the analytical solutions. Showing solutions to the first modal frequencies higher than the analytical ones.

Figure 5: Linear convergence for **AMC** element type

This discrepancy is likely related to the modal shapes obtained in these simulations. Visual inspection of the AMC mode shapes at low mesh densities reveals a significant deviation from the expected mode shapes, indicating that the non-conforming nature of AMC elements may lead to incorrect shape representation and consequently to high error margins for higher-order modes, fig. 7 & 8.

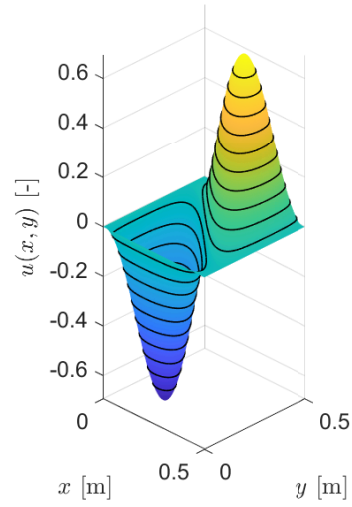
(a) Modal shape for 2×2 mesh grid (b) Modal shape for 16×16 mesh gridFigure 6: Modal shape for 1 – 1 natural frequencies and **AMC** elements type

Displacement Field from ACM Shape Functions
Solution for 2x2 elements mesh



(a) Modal shape for 2×2 mesh grid

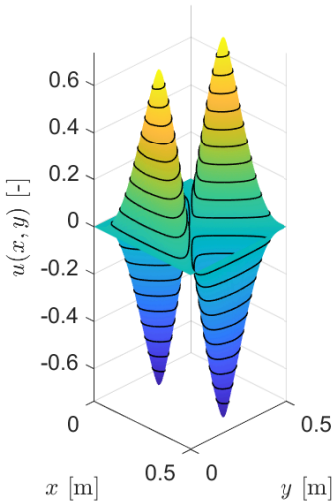
Displacement Field from ACM Shape Functions
Solution for 16x16 elements mesh



(b) Modal shape for 16×16 mesh grid

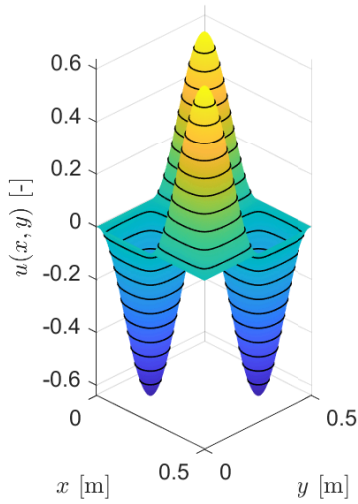
Figure 7: Modal shape for 1 – 2 & 2 – 1 symmetric natural frequencies and **AMC elements type**

Displacement Field from ACM Shape Functions
Solution for 2x2 elements mesh



(a) Modal shape for 2×2 mesh grid

Displacement Field from ACM Shape Functions
Solution for 16x16 elements mesh



(b) Modal shape for 16×16 mesh grid

Figure 8: Modal shape for 2 – 2 natural frequencies and **AMC elements type**

Note: previous images refer to **AMC** element type

Following images refer to **CR** element type

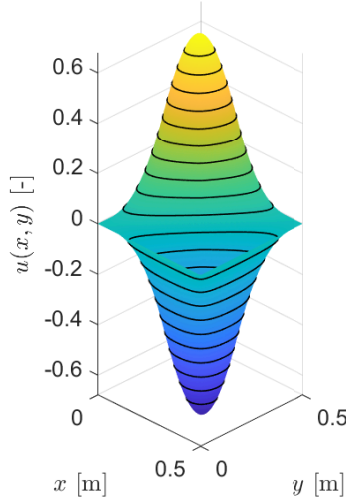
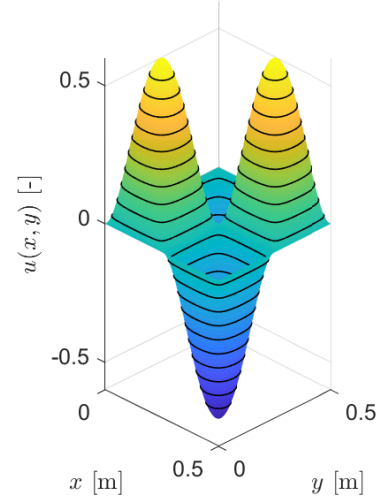
Displacement Field from CR Shape Functions
Solution for 2x2 elements mesh(a) Modal shape for $1 - 2/2 - 1$ modesDisplacement Field from CR Shape Functions
Solution for 2x2 elements mesh(b) Modal shape for $2 - 2$ modes

Figure 9: Modal shape for symmetric $1 - 2/2 - 1$ and $2 - 2$ natural frequencies for 2×2 mesh grid and **CR elements type**.

From visual analysis, a similar behavior can be observed even for the first mode shape. In particular, with a 2×2 element mesh, non-conformity between adjacent elements is clearly visible when using AMC elements, Figure 6. By plotting the first modal shape along the symmetry plane, Figure 11, it becomes evident that AMC elements provide a poorer approximation compared to CR elements, especially at low mesh densities.

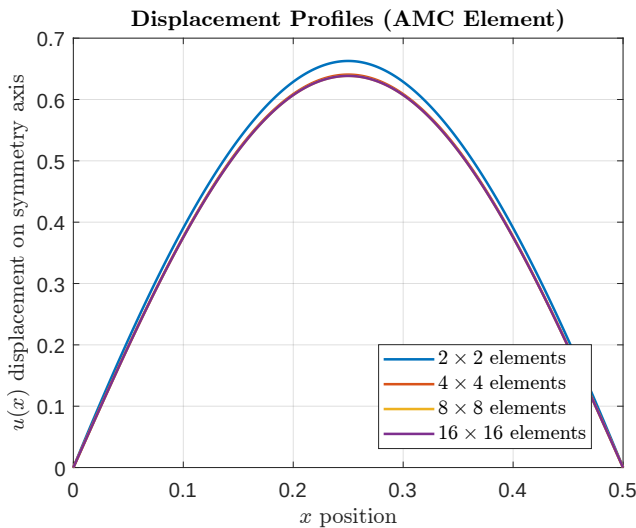
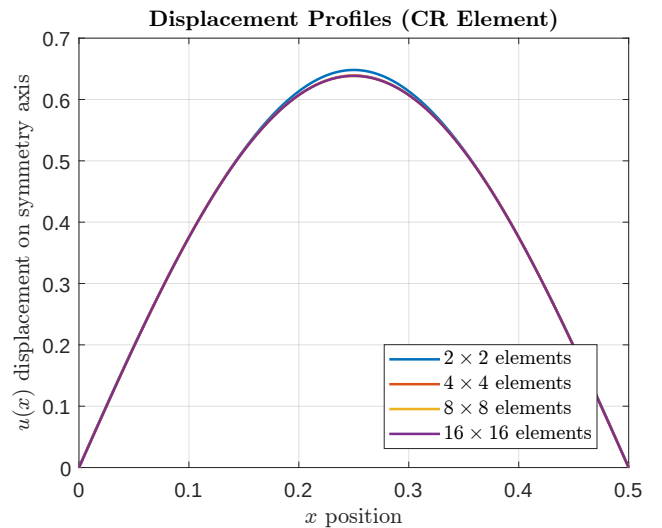
(a) Convergence for **AMC** 1st modal shape(b) Convergence for **CR** 1st modal shape

Figure 10

The displacement discrepancies with respect to the converged mode shape are more pronounced for AMC elements, and they tend to diminish more slowly as the mesh is refined. This observa-

tion is consistent with the previously discussed convergence trends, confirming the slower and less accurate convergence of AMC elements for the first mode.

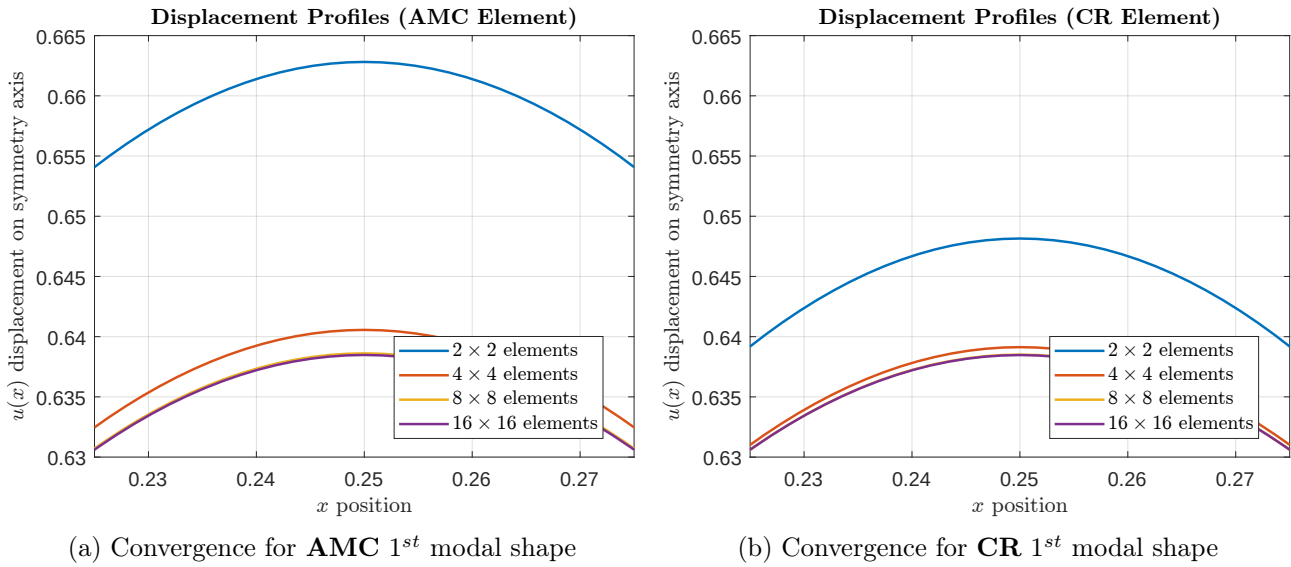


Figure 11

On the other hand, CR elements provide much more accurate results even for coarse meshes, Figure 9. For example, the 2×2 mesh already reproduces a closer approximation to the convergence mode shapes for the higher modes mentioned above, aligning well with the observed convergence trends and confirming the robustness of the CR formulation.

In conclusion, CR elements demonstrated that, despite their higher complexity and increased computational cost, the inclusion of an additional twist degree of freedom and the use of fully conforming shape functions enable a more accurate approximation of the Kirchhoff plate theory. This results in a higher convergence rate and a significant reduction in the need for fine mesh refinement to achieve accurate results.

Nevertheless, the use of AMC elements can still be justified in certain engineering contexts. For coarse mesh discretizations, they tend to underestimate the first modal frequencies compared to the analytical solution. This conservative estimation can be considered advantageous in preliminary design stages, as it introduces a more prudent safety margin into the structural analysis.

Discussion on Thick Plate Simulation Using Thin Plate Formulations

To further investigate the applicability of thin plate finite elements in modeling thick plates, additional simulations were conducted by increasing the thickness-to-span ratio from the nominal $h/\ell = 0.01$ (i.e., $h = 0.01$ m for $\ell = 1$ m) to a more substantial value of $h/\ell = 0.1$ (i.e., $h = 0.10$ m). These changes were set to increase deviations from the underlying assumptions of Kirchhoff-Love plates theory, on which both AMC and CR elements are based upon, in order to compare the error of this FEA with respect to higher-order analytical solutions provided by Mindlin-Reissner plate theory.

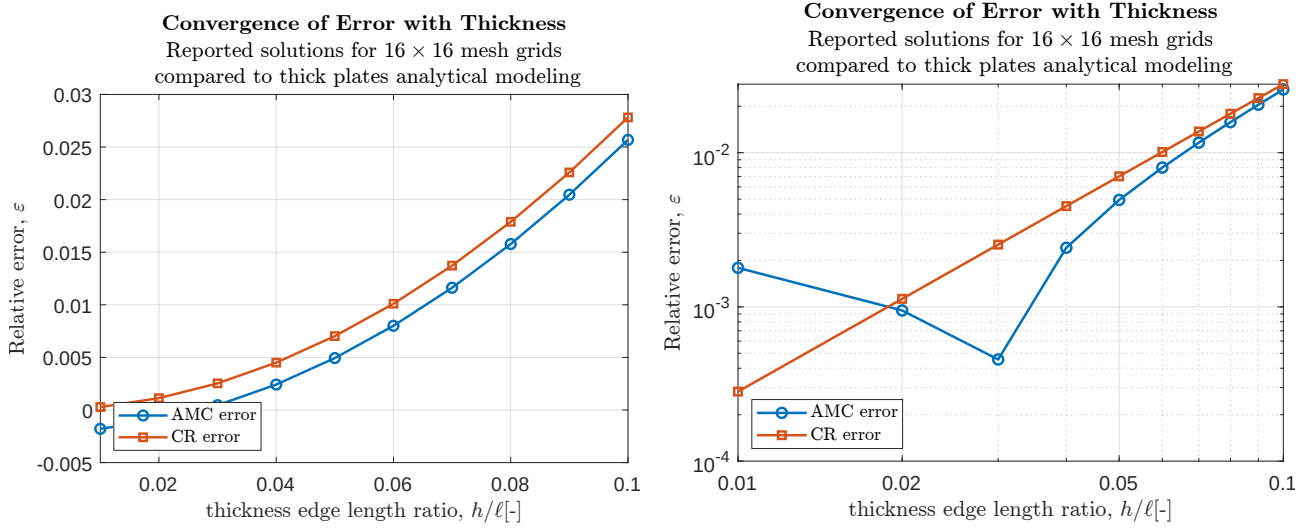


Figure 12: Error analysis on 1st modal frequencies at increase thickness to length ratio

The error analysis, reported in Figure 12, reveals distinct and interesting behaviours for the two element types. The relative error for the CR element in the first modal frequency, using a 16×16 mesh grid, shows a monotonically linear increase in a log-log plot as the plate thickness increases. This behaviour is consistent with the theoretical modeling: as the plate thickness grows, the assumption of negligible transverse shear deformation becomes increasingly invalid, causing the modelled system to exhibit a steadily increasing stiffness and, consequently, an overestimation of the first natural frequency.

In contrast, AMC elements exhibit a similar increase in stiffness of the modeled system, but because their convergence to the analytical solution is typically from below, as the thickness-to-length ratio increases, they inadvertently compensate for the missing shear deformation, achieving a closer convergence at intermediate values (ie $h/\ell = 0.03$). Yet, as the thickness grows further, the lack of a proper shear deformation mechanism in the formulation ultimately leads to a breakdown in accuracy.

These results highlight the limitations of applying classical thin plate elements to thick plate scenarios. For both element types, it can be observed that even at a relatively large thickness-to-length ratio of 0.1, the estimated values using a 16×16 mesh get to $\varepsilon \sim 2.5\%$. However, the accuracy quickly deteriorates as the thickness increases further. The behavior of the AMC elements, although seemingly advantageous at intermediate thicknesses, cannot be considered reliable for thicker plates due to the change in sign of the relative error, which contradicts the earlier assumption of convergence from below. In contrast, the monotonically increasing error of the CR elements may reflect a more predictable divergence from the theoretical model as the assumptions underlying the thin plate theory become invalid.

It is worth noting that, although the FEA results may deviate from the more accurate Mindlin-Reissner plate theory, they consistently exhibit a similar relative error when compared to the thin plate theory used to model these "thicker" plates (see Figure 13). This highlights the inherent limitations of the thin plate finite element formulations, especially as the validity of the underlying assumptions begins to break down.

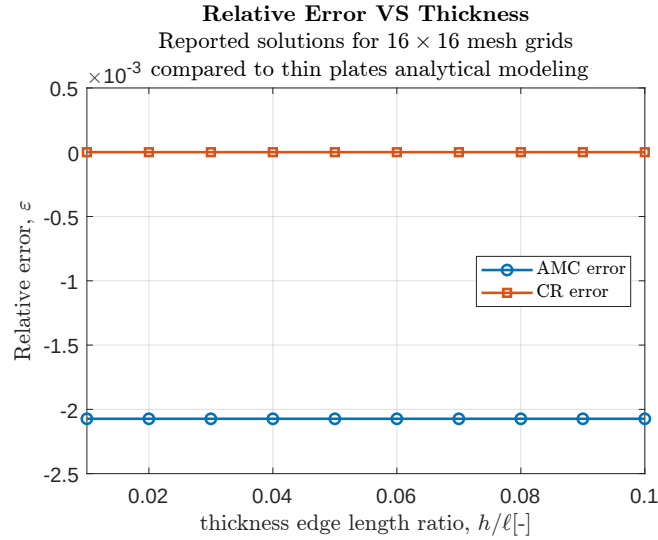


Figure 13: Relative error between FEA first modal frequency estimation and analytical thin plate solution in the thickness to length ratio of: $h/\ell \in [0.01, 0.1]$

Overall, these findings confirm that neither AMC nor CR thin plate elements are suitable for accurately modeling thick plates. For such cases, plate formulations based on Mindlin-Reissner theory or solid elements incorporating transverse shear deformation should be employed to ensure valid results.

5 Conclusion

In this study, we compared two plate element methods, Allman-McLeod-Clough (AMC) and Crouzeix-Raviart (CR), to predict the natural vibration frequencies of a simply supported square plate. Our findings can be summarized as follows:

- **Accuracy and Mesh Density:** CR elements achieved good accuracy even with coarse meshes (e.g., 2×2 grid), while AMC elements required finer meshes (around 8×8) to reach within 1% of reference frequencies. This means CR elements converge faster, with a convergence rate h^4 , and need fewer elements to give reliable results, while AMC convergence is h^2 .
- **Mode Shape Representation:** The CR formulation captured the correct shape of the vibrating plate much better at low mesh densities. AMC mode shapes tended to look distorted unless the mesh was refined further.
- **Effect of Plate Thickness:** As plate thickness increased, both AMC and CR results began to deviate from true frequencies. AMC elements showed a temporary compensation effect for shear deformation at moderate thickness, but neither element type is reliable for thick plates without accounting for shear deformation properly, as studied in class.
- **Computational Cost vs. Safety Margin:** AMC elements generally underestimate natural frequencies, which can act as an unplanned temporary 'safety cushion'. CR elements, although more accurate, introduce an additional degree of freedom per node, increasing computational effort.

Based on these results, we recommend the following:

- For thin plates where bending dominates, use CR elements to achieve fast convergence and accurate frequencies with minimal mesh refinement.
- For early design checks or when computational resources are limited, AMC elements can be used with the understanding that they may underestimate frequencies slightly, giving a conservative estimate.
- For thick plates or cases where shear deformation is important, consider using higher-order plate theories or full three-dimensional solid elements to capture shear effects accurately.

Overall, the choice between AMC and CR elements depends on the balance between accuracy, computational cost, and safety margin.

If based on the environment the plate operates in, a certain frequency response needs to be studied further, one could increase the node density at the 'peaks' and 'valleys' of the relevant mode shape, where the displacement is greatest, as we mentioned in the first mini project.

6 References

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